Copernicus Global Land Operations – biophysical variables (CGLOPS) Lot 2 “Hydrosphere and Cryosphere”

Framework Service Contract N° 945954 – IPR - 2023 (JRC)

QUALITY ASSESSMENT REPORT

SNOW COVER EXTENT

COLLECTION 1 KM GLOBAL

VERSION 1.0

Issue I1.00

Organization name of lead contractor for this deliverable: ENVEO IT GmbH

Book Captain: Gabriele Schwaizer (ENVEO)

Contributing Authors: Maria Heinrich, Nico Mölg, Thomas Nagler, Ursula Fasching, Andrea Scolamiero, Markus Hetzenecker, Johanna Nemec, Beat Haslinger (ENVEO)

Kari Luojus (FMI)

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

Document Release Sheet

Book captain:	Gabriele Schwaizer (ENVEO)	Sign Date
Approval:	Philippe Mourot (CLS)	Sign Date
Endorsement:	Nadine Gobron (JRC)	Sign Date
Distribution:	Public	

Change Record

Issue/Rev	Date	Page(s)	Description of Change	Release
	30.11.2025	All	First version	I1.00

© C-GLOPS Lot2 consortium

TABLE OF CONTENTS

1 Background of the document 11

1.1 Executive Summary 11

1.2 Scope and Objectives 11

1.3 Content of the document 12

1.4 Related documents 12

1.4.1 Applicable documents 12

1.4.2 Input 13

1.4.3 Output 13

1.4.4 External documents 13

2 Review of Users Requirements 15

3 Quality Assessment Methods 16

3.1 Reference data set 16

3.1.1 High resolution optical satellite data 16

3.1.2 Selection of reference scenes 17

3.1.3 Generation of reference snow maps 19

3.2 Validation procedure 20

3.3 Intercomparison procedure 22

4 Validation Results 24

4.1 Results For Global SCE Products from S3 SLSTR Data 24

4.2 Illustrative examples 27

4.3 Impact of cloud misclassification 29

5 Intercomparison Results 30

6 Conclusions 32

7 Recommendations 35

8 References 37

List of Figures

Figure 3.1: Temporal distribution of the 93 selected reference scenes selected for validation of the global SCE product from S3 SLSTR data over the NH. The same scenes are also used to validate the Northern Hemisphere SCE products from S-NPP VIIRS and S3 SLSTR data, enabling a direct comparison of the quality assessments of these products. 18

Figure 3.2: Spatial distribution of reference scenes for validating the global SCE product from S3 SLSTR data. Colors indicate the number of reference scenes acquired on different dates: 1 - purple, 2 - turquoise, 3 - yellow. The same scenes are also used to validate the Northern Hemisphere SCE products from S-NPP VIIRS and S3 SLSTR data, enabling a direct comparison of the quality assessments of these products. 18

Figure 3.3: Validation concepts adapted from the ESA QA4EO project SnowPEx (lead: ENVEO) for the validation of the global and NH SCE product with reference snow maps from high-resolution optical satellite data. 21

Figure 4.1: Box-whisker plot of bias and unbiased RMSD values from the validation of the global SCE product using 93 reference snow maps from the Northern Hemisphere acquired between 2020 and 2022. 24

Figure 4.2: Box-whisker plots of bias and unbiased RMSD values from validation of the NH SCE products from (a) S-NPP VIIRS data and (b) S3 SLSTR data, using the same 93 reference snow maps over the Northern Hemisphere from 2020 and 2022 as for validation summarized in Figure 4.1. 25

Figure 4.3: Bias values from the validation of the global SCE product from S3 SLSTR data against each reference snow map. 26

Figure 4.4: Unbiased RMSD values from the validation of the global SCE product from S3 SLSTR data against each reference snow map. 26

Figure 4.5: Illustrative validation example for the Landsat scene LC08_L1TP_095012_20210529_20210608_02, showing the data used in the validation and the spatial differences between the global SCE product from S3 SLSTR and the reference snow map (see text for detailed description of panels a – f). 27

Figure 4.6: Illustrative validation example for the Landsat scene LC08_L1TP_032035_20210112_20210308_02, showing the data used in the validation and the spatial differences between the global SCE product from S3 SLSTR and the reference snow map (see text for detailed description of panels a – f). 28

Figure 5.1: Quarterly aggregated difference maps between the global SCE product from archived S3 SLSTR data developed and generated in ESA CCI Snow (used for CLMS global SCE production from S3 SLSTR and NOAA-20 VIIRS data from December 2025 onwards) and the NH SCE product from S3 SLSTR data generated in NRT within CLMS for the period October

2020 – December 2022. Maps are organized according to the Northern Hemisphere hydrological cycle, starting with the snow onset season in October. 30

List of Tables

Table 3.1: Band numbers, wavelengths, and pixel spacing of Landsat 8/9 OLI/TIRS, and Sentinel-2 MSI. 16

Table 4.1: Mean, median, minimum and maximum bias and unbiased RMSD values for the global SCE product from S3 SLSTR and the NH SCE products from S-NPP VIIRS and S3 SLSTR data, validated using the same 93 reference snow maps. 25

Table 6.1: Compliance matrix for the global SCE product from S3 SLSTR data. ubRMSD = Unbiased Root Mean Square Difference. 33

List of Acronyms

ABBREV.	MEANING
ATBD	Algorithm Theoretical Basis Document
C3S	Copernicus Climate Change Service
CCI	Climate Change Initiative
CLMS	Copernicus Land Monitoring Service
ESA	European Space Agency
FSC	Fractional Snow Cover
GL	Global Land
IFOV	Instantaneous Field of View
JPSS-1	Joint Polar Satellite System-1
L1TP	Radiometrically and geometrically corrected Landsat data
MODIS	Moderate Resolution Imaging Spectroradiometer
MSI	Multi-Spectral Instrument
NDSI	Normalized Difference Snow Index
NDVI	Normalized Difference Vegetation Index
NFSF	Non-forested and sparsely forested
NH	Northern Hemisphere
NIR	Near Infrared
NOAA	National Oceanic and Atmospheric Administration
NRT	Near-Real Time
OLI	Operational Land Imager
QA4EO	Quality Assurance framework for Earth Observation
QAR	Quality Assessment Report
RMSD	Root Mean Square Difference
S3	Sentinel-3
S3A	Sentinel-3A
S3B	Sentinel-3B
SCE	Snow Cover Extent
SH	Southern Hemisphere
SLSTR	Sea and Land Surface Temperature Radiometer

© C-GLOPS Lot2 consortium

ABBREV.	MEANING
SnowPEx	Satellite Snow Product Intercomparison and Validation Exercise
S-NPP	Suomi National Polar-orbiting Partnership
SWIR	Shortwave Infrared
TIR	Thermal Infrared
TIRS	Thermal Infrared Sensor
ubRMSD	Unbiased Root Mean Square Difference
VIIRS	Visible Infrared Imaging Radiometer Suite
VIS	Visible
WMO	World Meteorological Organization

© C-GLOPS Lot2 consortium

1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Global Land (GL) Component in the framework of Copernicus is earmarked as a component of the Land service to operate “a multi-purpose service component” that will provide a series of bio-geophysical products on the status and evolution of land surface at global scale. Production and delivery of the parameters are to take place in a timely manner and are complemented by the constitution of long-term time series.

The Copernicus Global Land Service contains a Snow Cover Extent (SCE) Near Real Time (NRT) product covering global land areas excluding Antarctica at 0.01 degree pixel spacing. The global SCE product provides information on the Faction of Snow Cover (FSC) per pixel in percentage (=% - 100%). The SCE is derived from medium resolution optical satellite data.

The service is based on data from the Sentinel-3 (S3) Sea and Land Surface Temperature Radiometer (SLSTR) and the NOAA-20 (formerly Joint Polar Satellite System-1 (JPSS-1)) Visible Infrared Imaging Radiometer Suite (VIIRS). The global SCE products are generated in NRT as demonstration service from December 2025 onwards.

The spectral capabilities of the sensors are used in combination with a transmissivity map describing the transmissivity of the forest canopy, spectral reflectance maps representative for snow free ground conditions, and surface classification maps to detect the FSC per pixel in percentage. The usage of the transmissivity map allows also to detect the snow on ground also in forested areas.

This Quality Assessment Report (QAR) describes the results of the quality assessment of the global SCE product at 0.01 degree resolution from S3 SLSTR data with a limited reference data set available for the Northern Hemisphere.

S3 SLSTR data from Sentinel-3A (S3A) and Sentinel-3B (S3B) satellites are used for the product generation. For the generation of the global SCE products used for the validation activities, S3A and S3B SLSTR Level 1B data prepared with the processing baseline SL__L1_.004.04 were used [ED1].

1.2 SCOPE AND OBJECTIVES

The document presents the results of a limited quality assessment of the global SCE product from S3 SLSTR data. Four selected global SCE products from S3 SLSTR data acquired between June 2021 and September 2022 are considered for the quality assessment.

The quality assessment is performed on the global SCE product covering the land area of the Northern and Southern Hemisphere, excluding Antarctica, (90N/180W – 90S/180E) with 0.01 degree resolution using snow maps generated from high-resolution optical satellite data acquired nearly coincidently with the S3 SLSTR data.

The objective is to check that the overall product performance meets the quality requirements defined by users.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 reviews the users' requirements and the expected performance
- Chapter 3 describes the methodology for quality assessment, and the metrics and the criteria of evaluation
- Chapter 4 presents the results of the quality assessment analysis for S3 SLSTR-based global SCE products for the selected validation scenes
- Chapter 5 summarizes the main conclusions of the quality assessments
- Chapter 7 makes recommendations based upon the results

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: IGOS, 2007: Cryosphere Theme Report 2007. WMO/TD-No 1405. 100 pp. https://www.scar.org/library/scar-publications/occasional-publications/3511-integrated-global-observing-strategy-cryosphere-theme-report-for-the-monitoring-of-our-environment-from-space-and-from-earth/file/

AD2: GCOS, 2006: Systematic observation requirements for satellite-based products for climate. WMO/TD Report No. 1338, September 2006. https://library.wmo.int/doc_num.php?explnum_id=3813

AD3: GCOS – 200.2016. The Global Observing System for Climate: Implementation Needs. (GOOS-214) http://unfccc.int/files/science/workstreams/systematic_observation/application/pdf/gcos_ip_10oct2016.pdf

AD4: WMO OSCAR Observing Systems Capability Analysis and Review Tool; https://www.wmo-sat.info/oscar/observingrequirements

AD5: GCOS – 244.2022. The 2022 GCOS Implementation Plan. (GCOS-244) https://library.wmo.int/index.php?lvl=notice_display&id=22134.

AD6: GCOS – 245. 2025. The 2022 GCOS ECVs Requirements, 2022 edition - Updated in 2025. https://library.wmo.int/idurl/4/58111.

1.4.2 Input

Document ID	Descriptor
CGLOPS2_ATBD_SCE-GLOBE-1km_V1	Algorithm Theoretical Basis Document of the SCE-GLOBE-1km_V1 product
CGLOPS2_QAR_SCE-NHEMI-1km_V1	Product Quality Assessment Report providing validation activities and results for the SCE-NHEMI-1km_V1 product from S-NPP VIIRS and S3 SLSTR data, Issue 1.05
CGLOPS2_ATBD_SCE-NHEMI-1km-V1	Algorithm Theoretical Basis Document describing the method and algorithm as well as required input data to generate the SCE-NHEMI-1km V1 product based on VIIRS and SLSTR data, I1.04.

1.4.3 Output

Document ID	Descriptor
CGLOPS2_QAR_SCE_GLOBE-1km_V1	Product Quality Assessment Report providing validation activities and results for the SCE-GLOBE-1km_V1 product from S3 SLSTR data.

1.4.4 External documents

ED1: Optical Mission Performance Cluster 2022. Sentinel-3 Product Notice, S3.PN-SLSTR-L1.09. ESA, EUMETSAT. 31.05.2023. https://sentinel.esa.int/documents/247904/4755279/S3.PN-SLSTR-L1.09+-+i1r1+-+SLSTR+L1+PB+SL__L1_.004.04.00.pdf

ED2: CryoLand URD 2012. CryoLand User Requirement Document, URD - Final version. 31.05.2012. http://cryoland.enveo.at/docs/publications/CryoLand_No262925_D06.1_URD.pdf

ED3: ESA CCI Land Cover. 2017. Product User Guide 2.0. CCI-LC-PUG_V2, last modified: 31.03.2017. https://www.esa-landcover-cci.org/?q=webfm_send/112

ED4: Ripper, E., G. Schwaizer, T. Nagler, S. Metsämäki, M. Törmä, R. Fernandes, C. J. Crawford, T. Painter, K. Rittger. 2019. Guidelines for the generation of snow extent products from high-

resolution optical sensors. ESA QA4EO SnowPEx – The Satellite Snow Product Intercomparison and Evaluation Exercise, Deliverable D8, I2/1, pp.58. Link to PDF.

ED5: Bippus, G., T. Nagler, E. Ripper, R. Fernandes, S. Metsämäki. 2015. Methods and Protocols for intercomparing and validating snow extent and snow water equivalent products. ESA QA4EO SnowPEx – The Satellite Snow Product Intercomparison and Evaluation Exercise, Deliverable D7 – Snow Extent, I1/0, pp.50. Link to PDF.

2 REVIEW OF USERS REQUIREMENTS

According to the applicable documents [AD1] to [AD6], the external documents [ED2], Malnes et al. (2015) and Malenovský et al. (2012), the users' requirements relevant for the global Snow Cover Extent product are (T = Threshold, B = Breakthrough, G = Goal, as defined in the [AD6]):

- **Definition**: Global Snow Cover Extent (SCE)
- **Geometric properties:**
  - Spatial resolution:
    - T: 500 m² - 1000 m²
    - B: 10 m² - 500 m²
    - G: 10 m² - 100 m²
  - Grid / Projection: Geographic coordinates (latitude/longitude), WGS84 datum, WGS84 ellipsoid
- **Geographical coverage:**
  - Global land areas, excluding Antarctica, (90°N / 180°W – 90°S/180°E)
  - Target: Global land areas
- **Accuracy requirements:**
  - Geometric accuracy: Minimum: 1/3 IFOV, Target: IFOV 1 km (100 m in complex terrain)
  - Thematic measurement uncertainty (2-sigma):
    - T: 20%
    - B: 10 - 15 %
    - G: 5 %
- **Temporal requirements:**
  - Time period: Full year
  - Temporal frequency:
    - T: 48 hours - monthly
    - B: 24 hours
    - G: 3 - 6 hours
- **Delivery time:**
  - T: 240 hours after image acquisition - annually
  - B: 24 hours after image acquisition - weekly
  - G: 3 hours after image acquisition - daily

Note that user requirements are highly application dependent. Consequently, some of the specified requirements are more critical for near-real-time operations, while others are particularly relevant for climate applications.

3 QUALITY ASSESSMENT METHODS

The validation of the global SCE product is performed by comparing it with reference snow maps generated from high-resolution optical satellite data. The validation approach follows an updated version of the methods and protocols established within the European Space Agency (ESA) Quality Assurance framework for Earth Observation (QA4EO) Satellite Snow Products Intercomparison and Evaluation Exercise (SnowPEx), discussed and agreed by the international satellite snow community [ED4, ED5].

To assess the quality of the global SCE product, and to enable comparison with the existing Northern Hemisphere (NH) SCE products from Suomi National Polar-orbiting Partnership (S-NPP) VIIRS and S3 SLSTR data, the QAR currently focuses on validating the SCE product over the Northern Hemisphere. The same reference data set is used to validate both the global S3 SLSTR based SCE product and the NH SCE products from S-NPP VIIRS and S3 SLSTR data. Validation for the Southern Hemisphere (SH) will be incorporated during the demonstration service phase, beginning in December 2025, using reference snow maps from Landsat 8/9 and Sentinel-2.

3.1 REFERENCE DATA SET

3.1.1 High resolution optical satellite data

Archived high-resolution optical satellite data from the Landsat 8 and Landsat 9 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) are used to generate reference snow maps. The reference data base will be expanded in the future including imagery from the Sentinel-2 Multi-Spectral Instrument (MSI). The sensors have similar spectral band characteristics and spatial resolution, defined by their pixel spacing (Table 3.1). Thus, established methods and algorithms can be applied across these sensors with only minor adjustments.

Table 3.1: Band numbers, wavelengths, and pixel spacing of Landsat 8/9 OLI/TIRS, and Sentinel-2 MSI.

Spectral characteristic		*Landsat 8/9 OLI/TIRS*			*Sentinel-2 MSI*		
		Bd. No.	*Spectral band width (micro-metres)*	*Pixel spacing (meters)*	*Bd. No.*	*Spectral band width (micro-metres)*	*Pixel spacing (meters)*
Deep blue and violets (coastal blue, aerosol)		1	0.43 - 0.45	30	1	0.433 - 0.453	60
Visible (VIS)	Blue	2	0.45 - 0.51	30	2	0.458 - 0.523	10

Spectral characteristic		Landsat 8/9 OLI/TIRS			Sentinel-2 MSI		
		Bd. No.	Spectral band width (micro-metres)	Pixel spacing (meters)	Bd. No.	Spectral band width (micro-metres)	Pixel spacing (meters)
	Green	3	0.53 - 0.59	30	3	0.543 - 0.578	10
	Red	4	0.64 - 0.67	30	4	0.650 - 0.680	10
Near Infrared (NIR)					5	0.698 - 0.713	20
					6	0.773 - 0.793	20
					7	0.773 - 0.793	20
		5	0.85 - 0.88	30	8	0.785 - 0.900	10
					8A	0.855 - 0.875	20
					9	0.935 - 0.955	60
Shortwave Infrared (SWIR)		6	1.57 - 1.65	30	11	1.565 - 1.655	20
		7	2.11 - 2.29	30	12	2.100 - 2.280	20
Panchromatic (PAN)		8	0.50 - 0.68	15			
Cirrus / clouds		9	1.36 - 1.38	30	10	1.360 - 1.390	60
Thermal Infrared (TIR)		10	10.60 - 11.19	100* (30)			
		11	11.50 - 12.51	100* (30)			

* TIRS bands are acquired at 100 meters resolution but are resampled to 30 meters.

3.1.2 Selection of reference scenes

The reference scenes were selected based on the following criteria:

- The scenes have to be spatially and temporally representative.
- The scenes have to be cloud free or show only minor cloud coverage (< 5%).
- The scenes have to have a reasonably balanced distribution of snow and snow-free pixels.
- The scenes have to include a representative amount of open land.
- Each scene contained more than 300 pixels available for validation after the aggregation to the product grid spacing (0.01 deg x 0.01 deg).

No additional restrictions were applied during the selection of the reference scenes.

Based on these selection criteria, 93 reference scenes acquired over the NH between October 2020 and May 2022 were selected for validation.

Of these, 15 scenes are from the year 2020, 34 from 2021 and 32 from 2022 (Figure 3.1). The scenes are distributed across the NH land areas that are seasonally covered by snow (Figure 3.2).

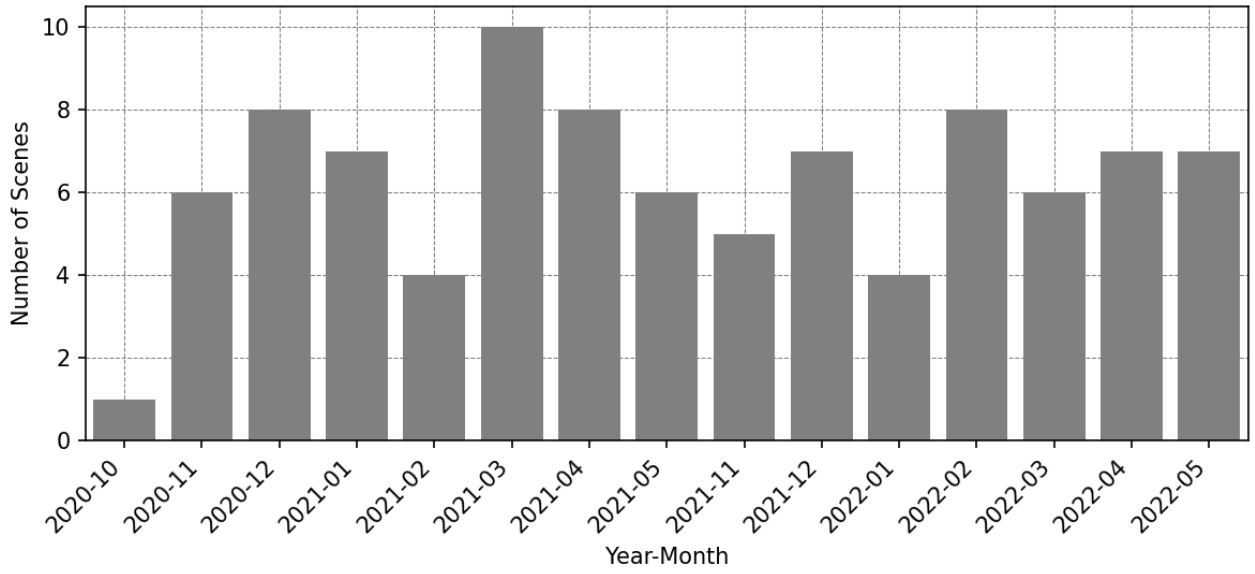


Figure 3.1: Temporal distribution of the 93 selected reference scenes selected for validation of the global SCE product from S3 SLSTR data over the NH. The same scenes are also used to validate the Northern Hemisphere SCE products from S-NPP VIIRS and S3 SLSTR data, enabling a direct comparison of the quality assessments of these products.

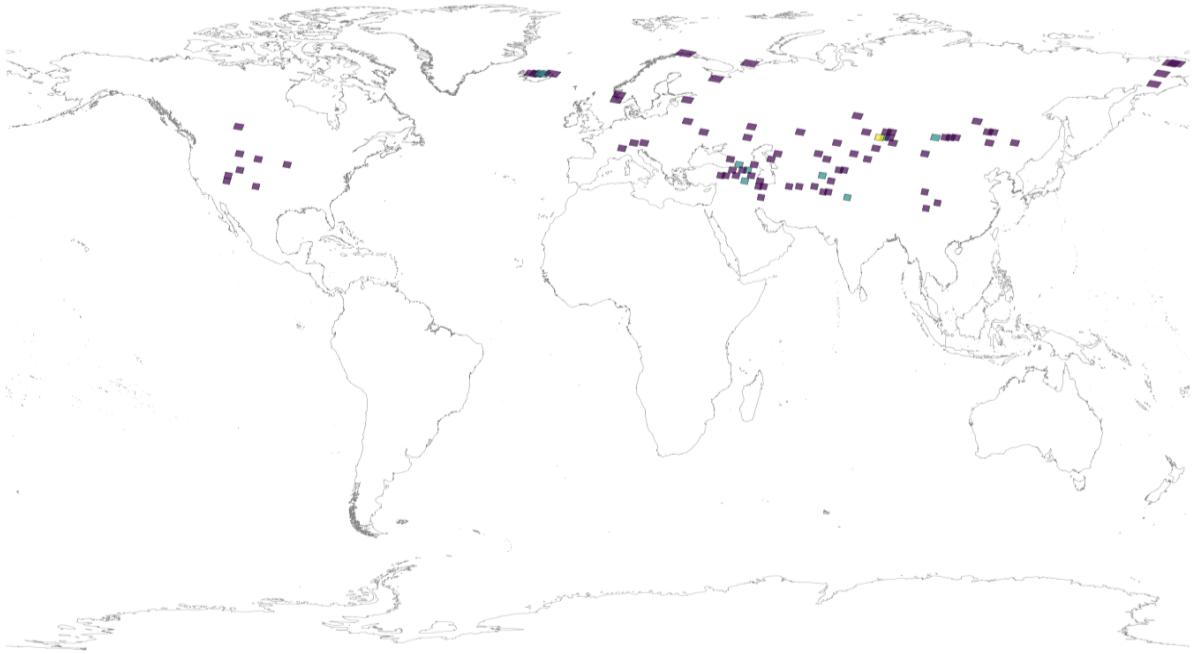


Figure 3.2: Spatial distribution of reference scenes for validating the global SCE product from S3 SLSTR data. Colors indicate the number of reference scenes acquired on different dates: 1 - purple, 2 - turquoise, 3 - yellow. The same scenes are also used to validate the Northern Hemisphere SCE products from S-NPP VIIRS and S3 SLSTR data, enabling a direct comparison of the quality assessments of these products.

3.1.3 Generation of reference snow maps

Reference snow maps are generated using a slightly adapted version of the approach applied in the previous validation of the NH SCE product, based on radiometrically and geometrically corrected Landsat Level1 (L1TP) products.

For this validation, Landsat 8 (LC08) and Landsat 9 (LC09) Collection 2 products (U.S. Geological Survey, 2021) are used, selected for their improved radiometric calibration and better alignment with Sentinel-2 data.

The Landsat L1TP data are radiometrically calibrated (Chander et al., 2009) and converted into spectral top of atmosphere reflectances and brightness temperatures. The original map projection of the Landsat images with 30 m pixel spacing is Universal Transverse Mercator (UTM) on the WGS84 ellipsoid. The data are bilinearly reprojected to geographic coordinates on the WGS84 ellipsoid with a pixel spacing of 0.00025 deg x 0.00025 deg.

Four snow detection algorithms are then applied to each Landsat scene:

- Dozier and Painter (2004): thresholds on the Normalized Difference Snow Index (NDSI) and a near-infrared (NIR) band, resulting in binary snow cover maps.
- Klein et al. (1998): thresholds on the Normalized Difference Vegetation Index (NDVI) and the NDSI, resulting in binary snow cover maps.
- Salomonson and Appel (2004, 2006): linear model weighting the NDSI, resulting in fractional snow cover maps.
- Keuris et al. (2023): multi-spectral unmixing approach, resulting in fractional snow cover maps.

Water bodies are masked using the Copernicus Global Surface Water (Pekel et al., 2016) product Maximum Water Extent (1984 – 2021), available in geographic coordinates on WGS84 ellipsoid with a grid spacing of 0.00025 deg x 0.00025 deg.

The resulting high-resolution snow maps from Landsat data are aggregated to the resolution of the SCE product, resulting in fractional snow cover (FSC) information for each reference map. During the aggregation, a pixel is included in the reference snow map only if at least 80% or its underlying data are observed.

A single reference snow map is then derived by computing the median value per pixel across the four aggregated FSC maps.

3.2 VALIDATION PROCEDURE

The global SCE product is generated daily from medium resolution optical satellite data with 0.01 deg x 0.01 deg pixel spacing in geographic coordinates on the WGS84 ellipsoid.

For the validation with selected reference data sets, archived S3 SLSTR data have been used for the generation of global snow map products. These products were generated within the ESA CCI Snow project. The applied algorithm [CGLOPS2_ATBD_SCE-GLOBE-1km_V1] detects the fraction of snow cover per pixel, expressed as a percentage from 0% and 100%. In forested areas, the algorithm accounts for the shading effect of the forest canopy, providing information on snow on the ground.

The validation approach is adapted from the SnowPEx protocol [ED4, ED5] and involves comparison with reference snow maps derived from high-resolution optical satellite data (cf. Section 3.1.3).

In the SnowPEx validation protocol, it is recommended calculating first the FSC in the original map projection and resolution of the high resolution satellite data and then reprojecting and aggregating the snow maps to the product grid. This approach is particularly useful when the same reference data set is intended for validating multiple SCE products with different map projections or resolutions.

In this validation, all products to be validated have the same grid specifications. Therefore, the high resolution spectral reflectance bands are first bilinearly reprojected to the product projection, and the FSC map is then calculated and aggregated to the product resolution.

The adjusted SnowPEx concept for product validation using reference snow maps from high-resolution optical satellite data is shown in Figure 3.3. The reference snow maps are generated using algorithms that do not account for shading effects from the forest canopy, unlike the algorithms used for the product generation. As a result, these reference maps represent the FSC viewable from above, while the products provide in forested areas a canopy corrected estimate of the FSC on the ground. Therefore, the validation is conducted in open land and sparsely forested areas based on ESA CCI land cover [ED3], as in these areas both thematic snow information, the snow cover on ground and the viewable snow cover, are the same. Densely forested regions are excluded from validation. Mountainous and plain areas are not treated separately in this validation.

The adapted SnowPEx approach consists of six major steps:

(1) Selection of high resolution (HR) optical satellite scenes,

(2) Reprojection of HR reflective and emissive bands to the product map projection,

(3) Generation of HR snow maps in the product map projection,

(4) Aggregation of HR snow maps to product resolution,

(5) Generation of a single reference snow map by computing the median of multiple aggregated snow maps,

(6) Pixel-based validation of the full sample of aggregated reference snow pixels in the SCE product.

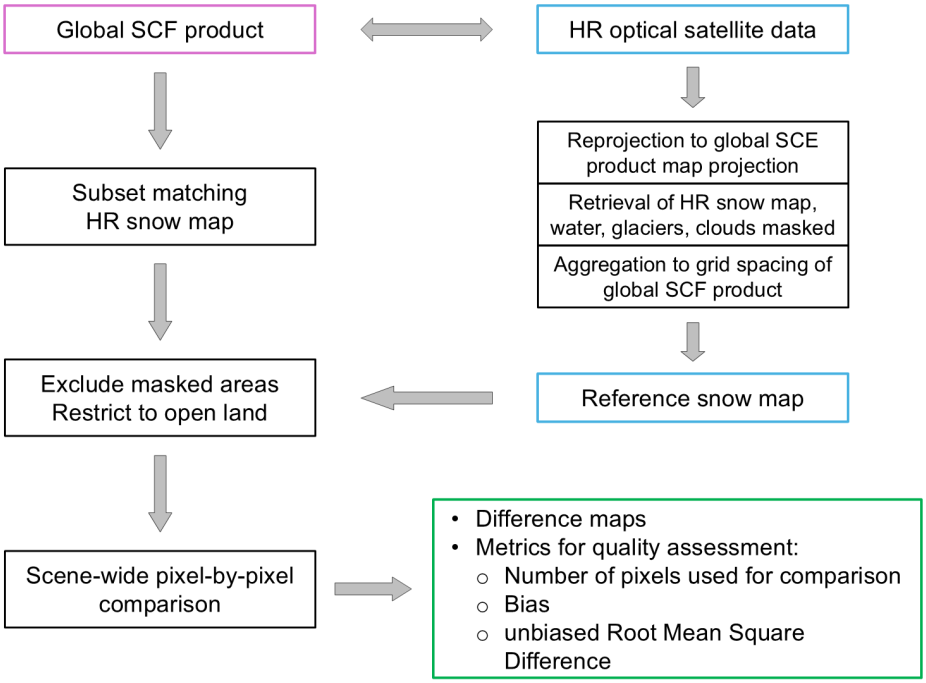


Figure 3.3: Validation concepts adapted from the ESA QA4EO project SnowPEx (lead: ENVEO) for the validation of the global and NH SCE product with reference snow maps from high-resolution optical satellite data.

Pixels classified as water bodies, clouds, or invalid pixel information in either the global SCE product, the NH SCE product, or the reference snow product, are excluded from the validation process. For all other areas, a pixel-by-pixel intercomparison is applied calculating the following statistical measures to describe the product performance:

- For determining the bias between two snow products, the number of valid pixels used for the inter-comparison (N_{ui}) is used as calculation basis, with the Fractional Snow Cover Extent per pixel ($\boldsymbol{FSC_{EXT}(i,j)}$) of the SCE product as well as the Fractional Snow Cover per pixel from the reference snow map ($\boldsymbol{FSC_{REF}(i,j)}$), taking into account all valid non-forest or

sparsely forested (NFSF) pixels, including snow-free pixels (FSC = 0 %) and all pixels covered partly or fully by snow (FSC = 1 % to 100 %):

$$BIAS = \frac{1}{N_{ui}} \sum_{j=0}^{y} \sum_{i=0}^{x} \left(FSC_{EXT}(i,j) - FSC_{REF}(i,j) \right) \quad \text{Eq. 3.1}$$

- The root-mean-square difference, RMSD, between two snow products is calculated using all pixels suitable for inter-comparison (N_{ui}) with Fractional Snow Cover Extent per pixel ($\boldsymbol{FSC_{EXT}(i,j)}$) of the SCE product as well as the Fractional Snow Cover per pixel from the reference snow map ($\boldsymbol{FSC_{REF}(i,j)}$), according to

$$RMSD = \sqrt{\frac{1}{N_{ui}} \sum_{j=0}^{y} \sum_{i=0}^{x} \left(FSC_{EXT}(i,j) - FSC_{REF}(i,j) \right)^2} \quad \text{Eq. 3.2}$$

- In addition to the RMSD, the unbiased RMSD is applied using the same input dataset (N_{ui}) with Fractional Snow Cover Extent per pixel ($\boldsymbol{FSC_{EXT}(i,j)}$) of the SCE product as well as the Fractional Snow Cover per pixel from the reference snow map ($\boldsymbol{FSC_{REF}(i,j)}$), and the average fractional snow cover value of each, the product ($\overline{FSC_{EXT}}$) and the reference snow map ($\overline{FSC_{REF}}$), respectively, and is described by

$$unbiased\ RMSD = \sqrt{\frac{1}{N_{ui}} \sum_{j=0}^{y} \sum_{i=0}^{x} \left(\left(FSC_{EXT}(i,j) - \overline{FSC_{EXT}} \right) - \left(FSC_{REF}(i,j) - \overline{FSC_{REF}} \right) \right)^2} \quad \text{Eq. 3.3}$$

3.3 INTERCOMPARISON PROCEDURE

The methodological approach for the SCE intercomparison primarily follows the ESA SnowPEx protocols [ED5], which have been reviewed and endorsed by the international satellite snow community. This approach has been adapted for the intercomparison of the global SCE products generated from S3 SLSTR data within the ESA CCI Snow project with the NH SCE products from S3 SLSTR data generated within the Copernicus Land Monitoring Service (CLMS). These products share a common setup, including map projection (WGS-84 geographical coordinate system; EPSG: 4326), pixel spacing (0.01 degree), pixel alignment and data type. For the intercomparison of global and NH SCE products, the spatial extent of the NH SCE products (25°N – 84°N) is used.

The method developed within ESA CCI Snow for generating daily global SCE products from medium resolution optical satellite data (Schwaizer et al., 2025) will be used to produce daily global SCE

products from S3 SLSTR and NOAA-20 VIIRS data in Near-Real-Time (NRT) within the CLMS, starting December 2025.

The intercomparison between products is performed on a daily, pixel by pixel basis. For this purpose, daily intersection masks are generated. For each comparison, all products must be available on a given day to create an intersection mask for that day. Only pixels that contain valid FSC values (0-100%) in all SCE products are considered in the daily intersection mask.

The SCE intercomparison is performed for the maximum common spatial extent of the products and uses daily global SCE product from S3 SLSTR data, and the NH SCE products from S3 SLSTR data for the period 1 October 2020 to 31 December 2022. Daily results are aggregated into seasonal values for analyses. Seasons are defined as winter (January-March, JFM), spring (April-June, AMJ), summer (July-September, JAS), and autumn (October-December, OND). For each season, spatial difference maps are generated to identify spatial variations. These maps are based on pixel-wise daily FSC differences between each two SCE products, which are then averaged to produce seasonal maps. When a season occurs multiple times in the analysis period, the seasonal map represents the average of daily differences across all years.

An intercomparison between the global SCE products from S3 SLSTR and from NOAA-20 VIIRS data will be performed during the demonstration service, starting in December 2025, and will be included in a future version of this document.

4 VALIDATION RESULTS

This chapter presents the validation results of the global SCE products from S3 SLSTR data. A total of 93 reference snow maps from high resolution optical satellite scenes spatially distributed across the NH, acquired between 2020 and 2022 were selected for the validation (section 3).

4.1 RESULTS FOR GLOBAL SCE PRODUCTS FROM S3 SLSTR DATA

Across the validation samples, the average bias is -0.29% FSC (median: -0.42 %), and the mean unbiased RMSD of 8.37% (median: 8.40 %) (see Table 4.1). These results indicate that the snow cover estimated by the global SCE product from S3 SLSTR data is generally very close to the snow cover classified in the reference snow maps. The boxplot (Figure 4.1) shows the distribution of bias and unbiased RMSD around their mean values as well as the presence of outliers. Negative bias values correspond to an underestimation of snow cover, while positive values indicate overestimation. Larger bias values are usually associated with higher unbiased RMSD values.

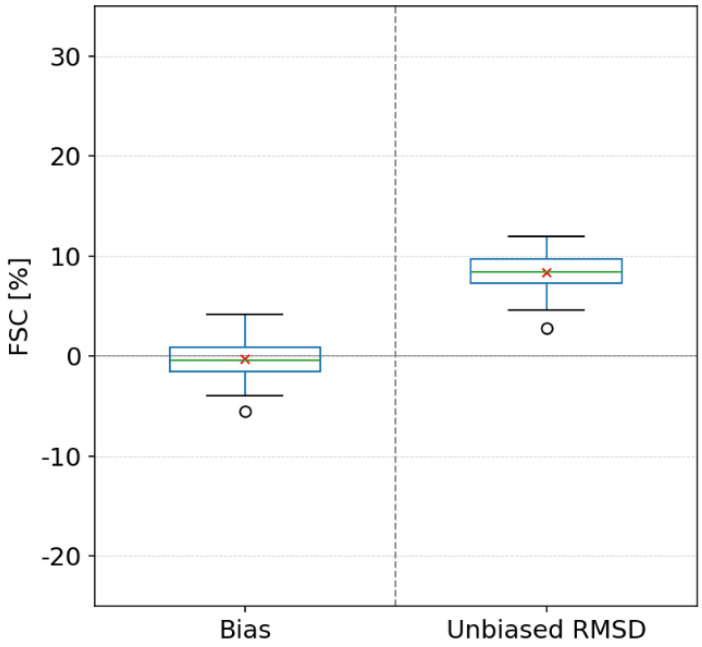


Figure 4.1: Box-whisker plot of bias and unbiased RMSD values from the validation of the global SCE product using 93 reference snow maps from the Northern Hemisphere acquired between 2020 and 2022.

For comparison, the NH SCE products from S-NPP VIIRS data and S3 SLSTR data, provided as NRT services within CLMS since 2018 and 2021, respectively, were evaluated using the same reference data set as the global SCE product from S3 SLSTR data. The results are summarized in the box-whisker plots shown in Figure 4.2. Compared with the NH SCE products from both sensors,

the global SCE products generated using the updated SCE retrieval method developed within ESA CCI Snow (Schwaizer et al., 2025) show reduced bias and lower unbiased RMSD values.

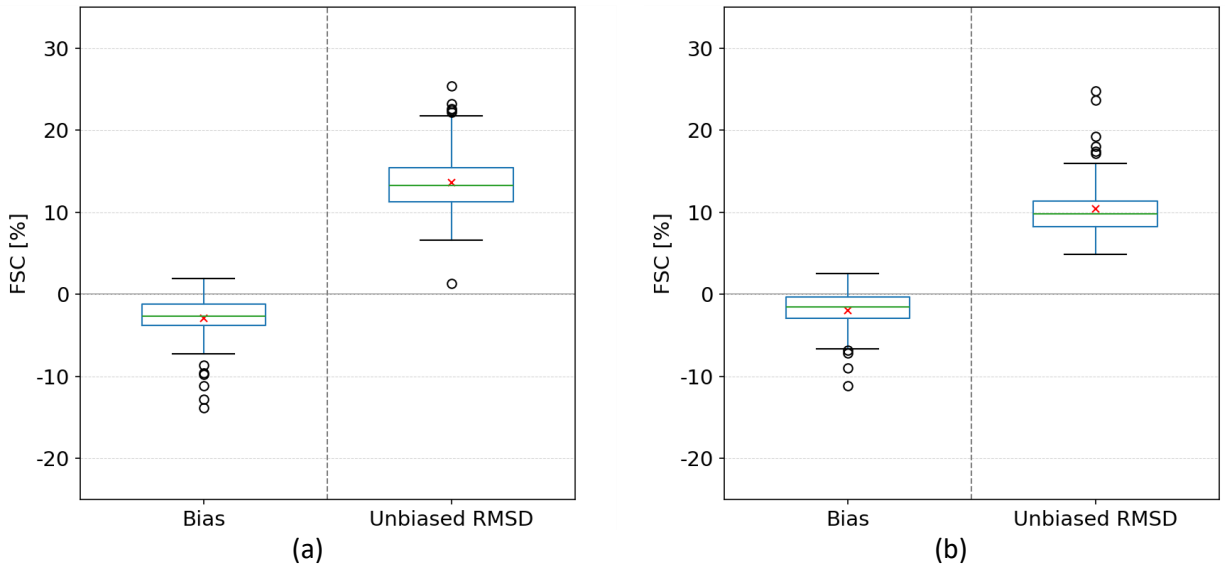


Figure 4.2: Box-whisker plots of bias and unbiased RMSD values from validation of the NH SCE products from (a) S-NPP VIIRS data and (b) S3 SLSTR data, using the same 93 reference snow maps over the Northern Hemisphere from 2020 and 2022 as for validation summarized in Figure 4.1.

The detailed validation values for the global SCE product from S3 SLSTR data, as well as for the NH SCE products from S-NPP VIIRS and S3 SLSTR data using the same reference data set, are summarized in Table 4.1

Table 4.1: Mean, median, minimum and maximum bias and unbiased RMSD values for the global SCE product from S3 SLSTR and the NH SCE products from S-NPP VIIRS and S3 SLSTR data, validated using the same 93 reference snow maps.

Metric	*Bias*			*Unbiased RMSD*		
Product	*GLOBAL SCE*	*NH SCE*	*NH SCE*	*GLOBAL SCE*	*NH SCE*	*NH SCE*
Satellite data base	*S3 SLSTR*	*S-NPP VIIRS*	*S3 SLSTR*	*S3 SLSTR*	*S-NPP VIIRS*	*S3 SLSTR*
Mean	-0.29	-2.96	-1.96	8.37	13.59	10.42
Median	-0.42	-2.62	-1.57	8.40	13.23	9.84
Minimum	-5.48	-13.83	-11.19	2.80	1.31	4.89
Maximum	4.22	1.94	2.54	12.01	25.36	24.76

For the global SCE product from S3 SLSTR data, the bias and unbiased RMSD values obtained from validation against each reference snow map are shown in Figure 4.3 and Figure 4.4, respectively. The bias values are generally close to zero. Larger deviations from zero are correlating with higher unbiased RMSD values and can be explained by patchy snow, which is difficult to detect with a medium-resolution satellite sensor. Additional details and illustrative examples are provided in section 4.2.

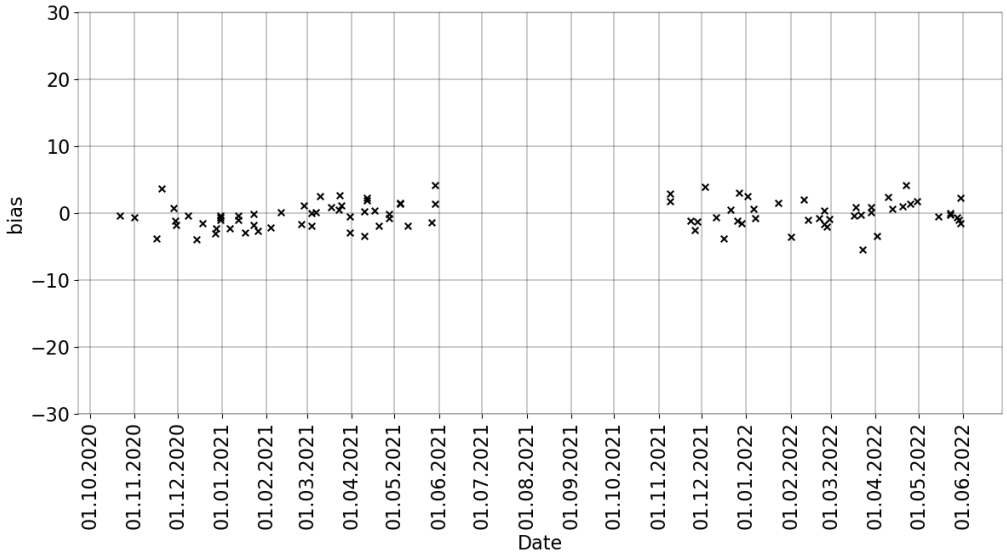


Figure 4.3: Bias values from the validation of the global SCE product from S3 SLSTR data against each reference snow map.

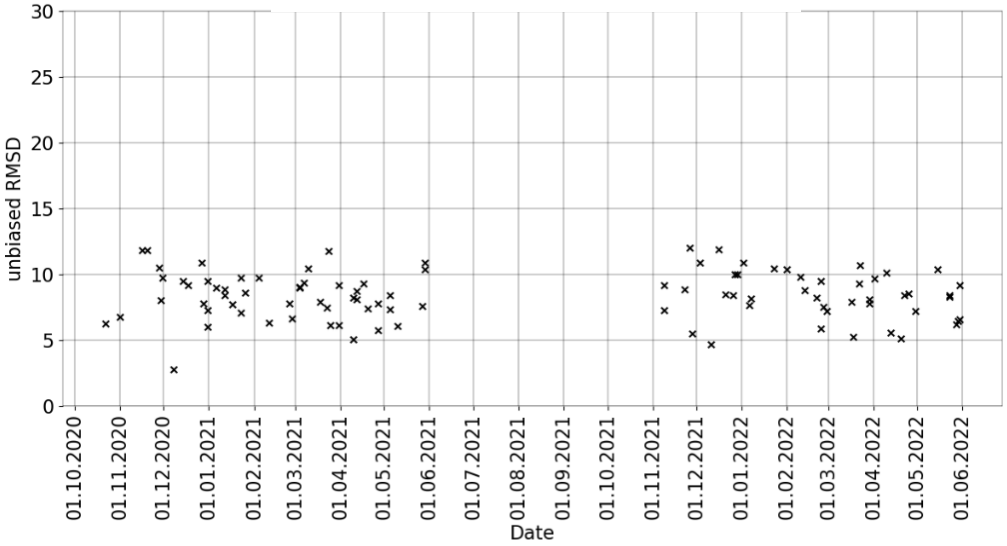


Figure 4.4: Unbiased RMSD values from the validation of the global SCE product from S3 SLSTR data against each reference snow map.

4.2 ILLUSTRATIVE EXAMPLES

Two days were selected to illustrate the validation of S3 SLSTR-based global SCE products using HR reference snow maps. Each example includes:

(a) The median reference snow map derived from four snow maps (see ED4, and Section 3.1.3).

(b) A false colour composite of the Landsat image (bands 6/5/4).

(c) The S3 SLSTR-based global SCE product for the Landsat scene.

(d) The difference map between the global SCE product and the reference snow map.

(e) The location of the reference snow map.

(f) The legends for the SCE maps and the difference map.

The first example in Figure 4.5 shows a scene in eastern Russia on 29 May 2021. Dense forest areas are masked in the snow maps and excluded from the validation. Both the Landsat scene and the S3 SLSTR-based global SCE product for this area are cloud-free.

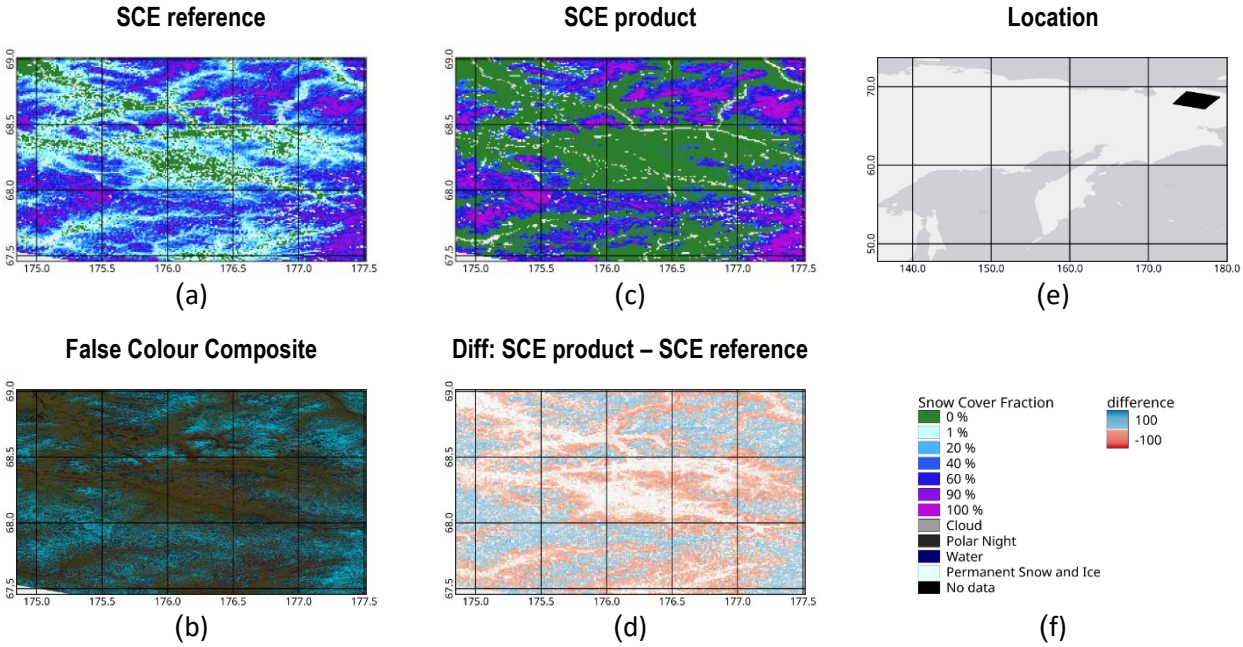


Figure 4.5: Illustrative validation example for the Landsat scene LC08_L1TP_095012_20210529_20210608_02, showing the data used in the validation and the spatial differences between the global SCE product from S3 SLSTR and the reference snow map (see text for detailed description of panels a – f).

Visual comparison between the median reference snow map from Landsat data and the global SCE product from S3 SLSTR data shows a high level of agreement, despite being one of the more challenging cases of the total of 93 validation cases. In this example, the bias of the global SCE product based on S3 SLSTR data compared to the reference snow map is -1,35 % and the unbiased RMSD is 10,87 %. The SLSTR-based global SCE product underestimates snow compared to the median reference snow map, particularly in areas with patchy snow cover.

In the second example, shown in Figure 4.6, the satellite data were acquired on 12 January 2021 over North America. As in the previous example, the S3 SLSTR-based global SCE product closely matches the reference snow map. Unlike the first scene, the snow cover in this scene is less patchy. This scene exhibits the second-lowest bias and unbiased RMSD among all reference snow maps, with a bias of -0.41% and an unbiased RMSD of 8.39%. Compared to the median reference snow map, the S3 SLSTR-based global SCE product slightly underestimates snow in patchy snow areas while slightly overestimating it in fully snow-covered regions.

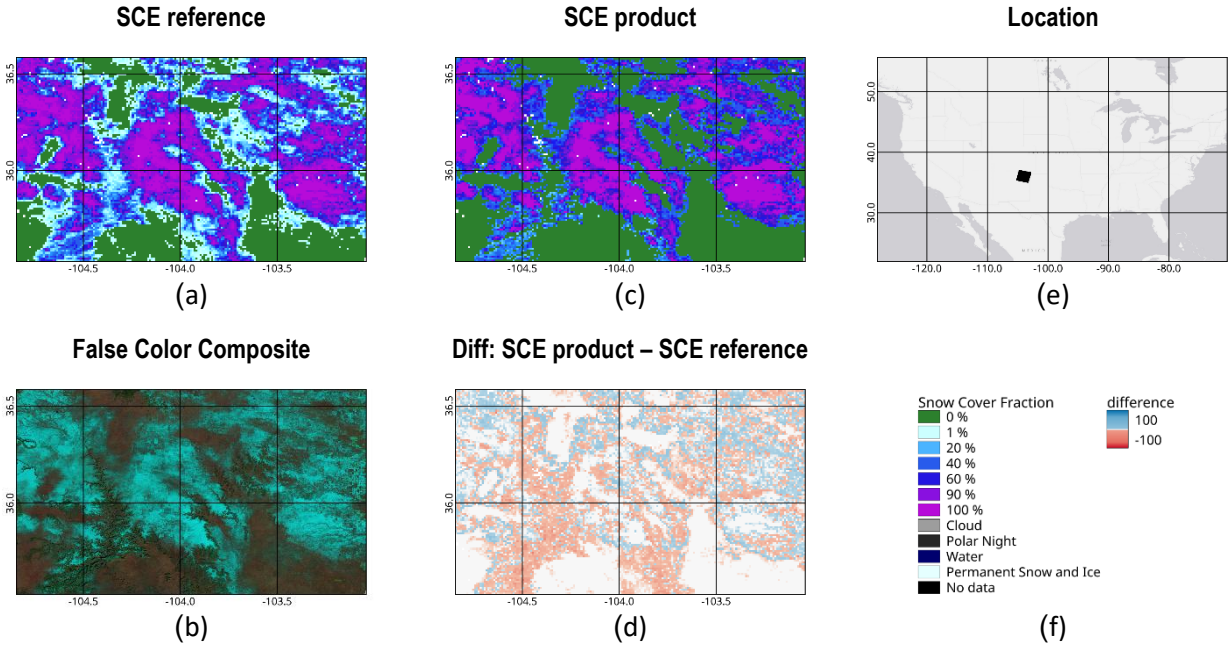


Figure 4.6: Illustrative validation example for the Landsat scene LC08_L1TP_032035_20210112_20210308_02, showing the data used in the validation and the spatial differences between the global SCE product from S3 SLSTR and the reference snow map (see text for detailed description of panels a – f).

Both examples illustrate the slight underestimation of snow in highly patchy snow areas. Conversely, pixels with higher FSC values tend to saturate more quickly, meaning that the reference snow map shows FSC values close to 100% while the global SCE product from S3 SLSTR data has already reached the maximum FSC value of 100%.

4.3 IMPACT OF CLOUD MISCLASSIFICATION

Cloud mapping in optical imagery remains challenging, particularly over snow and ice (e.g. Foga et al., 2017; Yin et al., 2022). Misclassification of clouds can significantly affect the quality of snow maps. Missed clouds are often erroneously classified as snow, while overly conservative cloud masks can obscure larges areas of land, resulting in reduced areas considered for the FSC classification in the snow map. This topic has been addressed in detail in the QAR of the NH SCE products (see CGLOPS2_QAR_SCE-NHEMI-1km_V1) and will be further investigated for the global SCE products and reported in a future version of this report.

For this limited quality assessment, reference scenes were selected with minimal cloud cover (<5%). However, the assumption that cloud-free conditions also prevailed at the time of the S3 SLSTR acquisitions may not always be valid. Consequently, the impact of clouds on the quality of the global SCE products is only marginally captured in this validation, due to the deliberate selection of nearly cloud-free Landsat scenes for generating the reference snow maps.

Accurate cloud detection over snow remains a major challenge with the spectral bands available on current optical sensors. Further research is needed to identify which spectral wavelengths and band widths could improve the discrimination between clouds and snow.

5 INTERCOMPARISON RESULTS

The global SCE product derived from S3 SLSTR data using the improved retrieval method developed within ESA CCI Snow (Schwaizer et al., 2025) was compared with the S3 SLSTR-based NH SCE product currently provided within CLMS (cf. Section 0). Spatial difference maps for the four main seasons (Figure 5.1) indicate that, overall, the global SCE product with the update SCE retrieval method detects more snow. Exceptions occur in northeastern Russia during the onset season (October to December, OND), where significantly less snow is observed, with slight underestimation persisting into the main winter (January to March, JFM) and spring (April to June, AMJ) seasons.

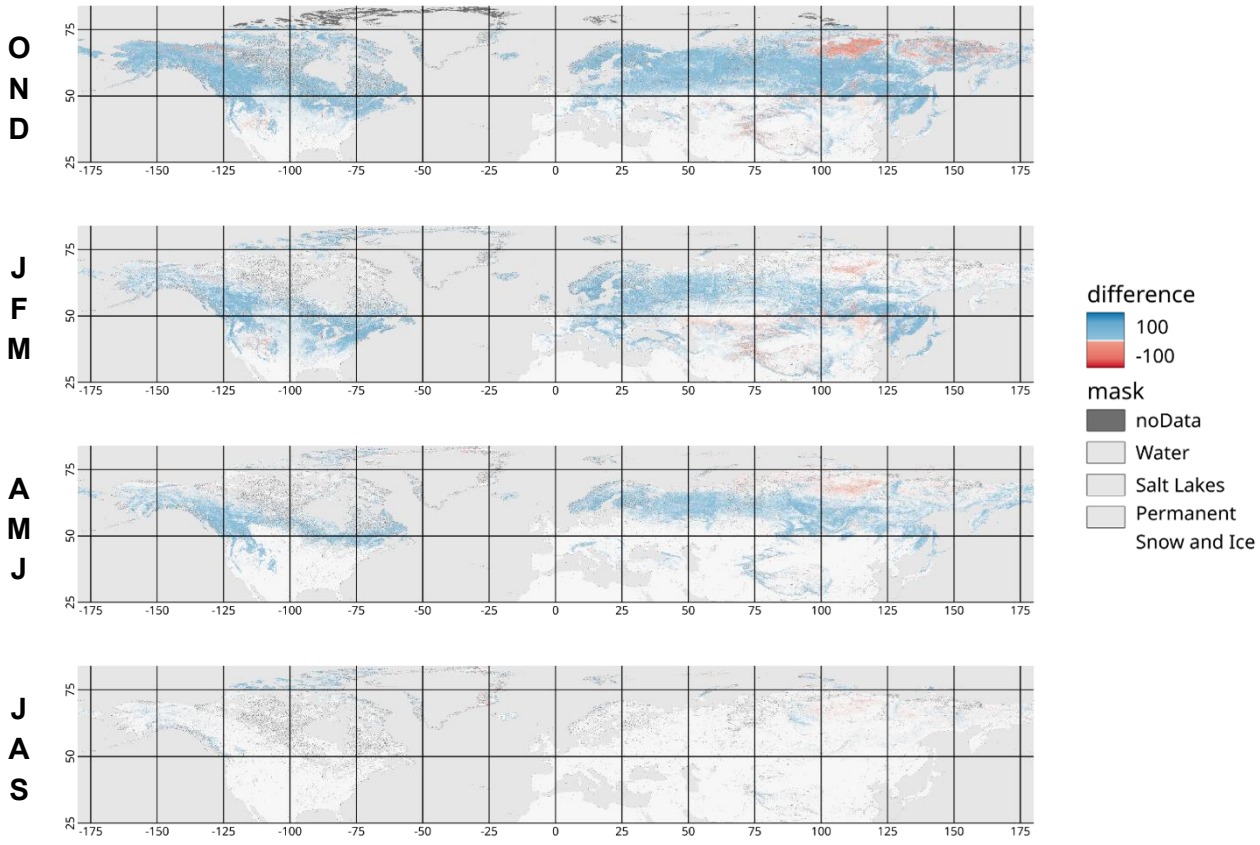


Figure 5.1: Quarterly aggregated difference maps between the global SCE product from archived S3 SLSTR data developed and generated in ESA CCI Snow (used for CLMS global SCE production from S3 SLSTR and NOAA-20 VIIRS data from December 2025 onwards) and the NH SCE product from S3 SLSTR data generated in NRT within CLMS for the period October 2020 – December 2022. Maps are organized according to the Northern Hemisphere hydrological cycle, starting with the snow onset season in October.

In the main winter season (JFM), some regiones in northern Canada, western North America and central Asia also show slightly less snow. During the melt season (AMJ), some of these differences diminish or even reverse, while differences during the summer season (July to September, JAS) are minimal.

Seasonally, the largest area-wise differences occur in the onset season (OND), followed by the main winter season (JFM) and melt season (AMJ). The updated SCE retrieval method improves detection of fully snow-covered areas, and in general, the global SCE product exhibits higher FSC values than the NH SCE product. However, patchy snow areas, particularly during the onset and melt seasons, remain challenging, and the estimation of FSC in these regions is still subject to uncertainties.

The intercomparison results are consistent with the validation results presented in Section 4, which showed that the updated SCE retrieval method generally detects more snow and reduces both bias and unbiased RMSD compared to reference snow maps from high resolution optical satellite data. The spatial and seasonal patterns observed from the intercomparison align with the improvements in snow detection accuracy documented during the validation.

© C-GLOPS Lot2 consortium

6 CONCLUSIONS

The quality of global SCE products derived from S3 SLSTR data was assessed using snow maps from high-resolution (HR) optical satellite data as reference. The validation covers the autumn, winter and spring seasons of 2020 to 2022 over the Northern Hemisphere, based on 93 selected Landsat 8 scenes. These reference data are spatially distributed across diverse surface types, snow and illumination conditions. Validation was performed on a pixel-by-pixel basis using statistical measures to quantify the performance of the global SCE product. The validation followed an adapted version of the validation protocol developed within the ESA QA4EO project SnowPEx+, which has been endorsed by the international satellite snow community [ED5].

Since the global SCE product is planned to replace the current NH SCE service in future, the NH SCE products from S-NPP VIIRS and S3 SLSTR data generated within CLMS were validated using the same reference data set. Results of this product quality assessment are summarized in this QAR. A more detailed and extensive quality assessment of the NH SCE products from both sensors over the Northern Hemisphere is provided in the QAR for the NH SCE products (CGLOPS2_QAR_SCE-NHEMI-1km_V1, I1.05).

Although high-resolution reference snow maps provide more accurate snow information, they still contain uncertainties, even in non-forested and sparsely forested areas. To minimize algorithm-specific biases, reference snow maps per Landsat scene were generated using four different SCE retrieval algorithms, which were then combined into a single reference snow map for each Landsat scene (see Chapter 3 for details).

Overall, the validation results demonstrate that the global SCE product from S3 SLSTR data meets the accuracy requirements defined by users (Table 6.1). For non-forested and sparsely forested areas, the mean unbiased Root Mean Square Difference (ubRMSE) is 8.37% (median: 8.40%), and the mean bias is –0.29% (median: –0.42%). This indicates that the global SCE product closely matches the reference snow maps. However, the classification of patchy snow areas from medium resolution optical satellite data remains challenging and such areas tend to be underestimated in the global SCE products.

Intercomparison of the global SCE product from S3 SLSTR data with the S3 SLSTR-based NH SCE product from CLMS shows that the updated retrieval method improves detection of fully snow-covered areas, generally producing higher FSC values in the global SCE product. Patchy snow areas, particularly during the onset and melt seasons, remain difficult to detect, but the updated product provides improved snow detection across most regions and seasons. Seasonally, the largest differences occur during the onset season, followed by the main winter and melt seasons, while

differences in the summer season are minimal. These findings are consistent with the validation results and demonstrate the enhancements introduced by the updated SCE retrieval method (Schwaizer et al., 2025).

A comprehensive validation for the Southern Hemisphere is still required. This will involve expanding the number and spatial distribution of reference samples. In addition, global SCE products derived from NOAA-20 VIIRS data, to be generated and provided as a backup near-real-time service, also need to be evaluated. The validation will be extended during the demonstration service alongside ongoing data provision, starting in December 2025. This activity will serve as a preparatory step towards a (pre-)operational near-real-time service in future.

Table 6.1: Compliance matrix for the global SCE product from S3 SLSTR data. ubRMSD = Unbiased Root Mean Square Difference.

Requirements	*Minimum*	*Target*	*Value*	*Range (Min/Max)*	*Compliant*	*Comments*
Spatial coverage:	Northern Hemisphere land areas	Global land areas	Global land areas	180°W / 90°N - 180°E / 90°S	Yes	
Spatial resolution:	500 m (mountains) - 1000 m (plains)	100 m, 10 m in complex terrain	0.01 deg x 0.01 deg (~ 1 km x 1 km)		Yes	
Thematic accuracy:	20 %	5 %	ubRMSD: 8.37 % / 8.40 % Mean/Median Bias: -0.29 % / -0.42 % Mean/Median	ubRMSD: 2.80 % / 12.01% Min/Max Bias: -5.48 % / 4.22 % Min/Max	Yes	The current results are only valid for the NH. Validation for the SH will be added during the demonstration service of the global SCE product. Reference snow maps need also validation, general performance is similar for different surface classes, but outliers can occur. High variability with snow conditions, can be caused by global SCE product or reference snow information.
Time period:	Full year	Full year	Full year	Demonstration service from	Yes	Daily global SCE products from archived

Requirements	*Minimum*	*Target*	*Value*	*Range (Min/Max)*	*Compliant*	*Comments*
				December 2025 onwards.		S3 SLSTR data for 09/2020 – 12/2022 were generated within the ESA CCI Snow project. From Jan 2023 onwards, daily global SCE products from S3 SLSTR data are generated within the C3S for Cryosphere with a latency time of 6 to 9 months. Permanent snow and ice areas in these products are globally masked.
Temporal frequency:	48 hours - monthly	3 - 6 hours	Daily		Yes	
Delivery time:	240 hours - annually	3 - 24 hours after image acquisition	24 hours	-	Yes	24 hours after the last image acquisition required to generated daily SCE maps

7 RECOMMENDATIONS

The results of the current quality assessment of global SCE products from S3 SLSTR data at 0.01 degree resolution, generated with an updated SCE retrieval algorithm and validated using a limited set of reference snow maps from high-resolution optical satellite data, indicate that the thematic accuracy requirement defined by users can be met. However, this assessment is based on only 93 reference snow maps over the Northern Hemisphere, and the results should therefore be considered preliminary. Greater variability is expected when more samples are included, and a truly global extent is considered.

An expanded validation is needed, incorporating both a wider temporal and spatial coverage of reference snow maps, including the Southern Hemisphere, is needed to provide more robust and reliable results. In addition, global SCE products derived from NOAA-20 VIIRS data need to be included in the extended validation. Intercomparison of the S3 SLSTR and NOAA-20 VIIRS based global SCE products will further support confidence in their quality and consistency when used together.

Results of the quality assessment for the NH SCE products from S-NPP VIIRS and S3 SLSTR data are documented in the QAR for the NH SCE product (CGLOPS2_QAR_SCE-NHEMI-1km_V1, I1.05). When validated with the same reference dataset, the updated SCE retrieval algorithm applied to S3 SLSTR data reduces the mean unbiased RMSD by approximately 2% and improves the mean bias from about -2% to nearly zero.

When interpreting the quality assessment of the global SCE product, it should be kept in mind that even snow maps from (very) high-resolution optical snow maps may contain misclassifications. The following points should be considered when generating reference snow maps from high resolution optical satellite data:

1. For the generation of a reference snow map from high-resolution optical satellite data, a snow mapping algorithm designed particularly for the spectral capabilities of the satellite data should be used. The use of a multi-spectral unmixing algorithm (e.g. Keuris et al., 2023) is a step towards sensor-tailored approaches.
2. In is mandatory to check each reference snow map for plausibility. Any maps containing clearly unrealistic or erroneous snow patterns must be excluded to ensure the reliability of the reference dataset used for the product validation.
3. The reference snow maps themselves need to be validated. Snow cover is highly dynamic and can change from fully snow covered to snow-free within a few hours. Consequently, the

availability of temporally and spatially matching reference data, such as from Landsat or Sentinel-2 scenes, is limited. Generating reliable reference maps is therefore challenging and requires careful planning and preparation.

8 REFERENCES

Dozier, J., Painter, T.H., 2004. Multispectral and hyperspectral remote sensing of alpine snow properties. Annu. Rev. Earth Planet. Sci. 32, 465–494. https://doi.org/10.1146/annurev.earth.32.101802.120404.

Foga, S., Scaramuzza, P.L., Guo, S., Zhu, Z., Dilley, R.D., Beckmann, T., Schmidt, G.L., Dwyer, J.L., Joseph Hughes, M., Laue, B., 2017. Cloud detection algorithm comparison and validation for operational Landsat data products. Remote Sens. Environ. 194, 379–390. https://doi.org/10.1016/j.rse.2017.03.026.

Keuris, L., Hetzenecker, M., Nagler, T., Mölg, N., Schwaizer, G., 2023. An Adaptive Method for the Estimation of Snow-Covered Fraction with Error Propagation for Applications from Local to Global Scales. Remote Sens. 15, 1231-1256. https://doi.org/10.3390/rs15051231.

Klein, A. G., Hall, D. K., & Riggs, G. A. (1998). Improving snow cover mapping in forests through the use of a canopy reflectance model. Hydrological Processes, 12, 1723–1744.

Malnes, E., Buanes, A., Nagler, T., Bippus, G., Gustafsson, D., Schiller, C., Metsämäki, S., Pulliainen, J., Luojus, K., Larsen, H.E., Solberg, R., Diamandi, A., Wiesmann, A., 2015. User requirements for the snow and land ice services – CryoLand. The Cryosphere. 9, 1191–1202. https://doi.org/10.5194/tc-9-1191-2015.

Malenovský, Z., Rott, H., Cihlar, J., Schaepman, M. E., García-Santos, G., Fernandes, R., Berger, M. 2012. Sentinels for Science: Potential of Sentinel-1, -2, and -3 missions for scientific observations of ocean, cryosphere, and land. Remote Sensing of Environment. 120, 91-101. https://doi.org/10.1016/j.rse.2011.09.026.

Pekel, J. F., A. Cottam, N. Gorelick, A. S. Belward. 2016. High-resolution mapping of global surface water and its long-term changes. Nature, 540, 418-422. doi:10.1038/nature20584.

Salomonson, V.V., Appel, I., 2006. Development of the Aqua MODIS NDSI fractional snow cover algorithm and validation results. IEEE Trans. Geosci. Remote Sens. 44, 1747–1756. https://doi.org/10.1109/TGRS.2006.876029.

Salomonson, V.V., Appel, I., 2004. Estimating fractional snow cover from MODIS using the normalized difference snow index. Remote Sens. Environ. 89, 351–360. https://doi.org/10.1016/j.rse.2003.10.016.

Schwaizer, G., S. Metsämäki, M. Moisander, K. Luojus, M. Heinrich, S. Wunderle, T. Nagler, J. Lemmetyinen, J. Pulliainen, M. Takala, R. Solberg, P. Venäläinen, L. Zschenderlein, N. Mölg, X. Xiao. 2025. ESA CCI+ Snow ECV: Algorithm Theoretical Basis Document, version 5.1, November 2025.

Yin, M., Wang, P., Ni, C., Hao, W., 2022. Cloud and snow detection of remote sensing images based on improved Unet3+. Sci. Rep. 12, 14415. https://doi.org/10.1038/s41598-022-18812-6.

U.S. Geological Survey, 2021. Landsat Collection 2 (ver. 1.1, January 15, 2021): U.S. Geological Survey Fact Sheet 2021–3002, 4 p. https://doi.org/10.3133/fs20213002.